***** WRITE YOUR NAME AND COHORT BELOW *****

UNIVERSITY OF CALIFORNIA BERKELEY
Haas School of Business

MBA 201A—Economic Analysis for Business Decisions—Fall 2018

Professor Steve Tadelis

MIDTERM EXAM

Instructions: The number in brackets (*e.g.,* [5]) indicates the points for each question. Total: 85 points. Note that you have 90 minutes to do the exam, so if you spend no more than 1 minute per point, you will have some time to check your work.

Please Write Legibly. Briefly explain your answers (that is, don't just write "yes" or "no" and don't just write down a numerical answer without showing how you derived it). Write only on this exam and not on other sheets of paper. The last page is a blank sheet for your use.

****** Please sign the honor code oath at the bottom of this page. ******

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Honor Code:

I understand that this exam is an individual exercise. I swear on my honor that I have not consulted with another person or made use of notes or other materials on- or off-line during the exam.

Name:_____

Cohort:_____

Short answer questions [20 points]

These 3 questions require **short answers** (2-5 sentences). Use any graphs or calculations that will help your explanation. Be sure to label graphs clearly.

1. [5] The Microsoft Surface Go is a 10-inch Windows-based tablet computer that was released on August 2nd, 2018. It can serve as a hybrid between a tablet and a lightweight laptop. Explain how much, if at all, this product might affect the demand for (1) iPads; and for (2) Microsoft cell phones?

2. [5] Your friend lives in mid-town Manhattan and pays $4 for a vanilla ice cream cone with rainbow sprinkles at the Mr. Softee Ice cream truck. After moving from mid-town to Brooklyn to save on rent, he learned that Mr. Softee trucks sell the same cone for only $2.50 there. He explains that costs are higher in mid-town because of the more expensive vending license needed to operate in Manhattan. Assume that vending licenses really are more expensive in Manhattan than in Brooklyn. How should an owner/operator of a Mr. Softee truck take the license fee into account when setting ice cream prices?

3. [10] The Elmwood Café has noticed a decline in the number of patrons having breakfast over the past month. They are debating changing their hours of operation and only serving lunch and dinner. They've called you in to advise them on their decision. They show you the following table:

Monthly Accounting at Elmwood, August 2015.
[Numbers in parentheses represent negative amounts.]

	Breakfast	Lunch	Dinner	Total
Hrs of operation	7AM-11AM	11AM-3PM	3PM-7PM	7AM-7PM
Meals served	1,200	3,700	1,500	5,900
Revenue	$3,600	$16,000	$7,500	$27,100
Food cost	$300	$4,000	$1,500	$5,800
Other direct costs*	$2,400	$4,800	$3,600	$10,800
Overhead**	$2,300	$2,300	$2,600	$6,900
Profit	($1,400)	$4,900	$100	$3,600

* Other direct costs include the wages of the counter and kitchen staff. All workers earn $15/hour. Elmwood staffs 2 people between 7AM and 11AM, 4 people from 11AM 3PM and 3 from 3PM-7PM. Elmwood is only open weekdays, or 20 days per month.

** Overhead includes the lease on the building, the manager's salary, and licensing fees that they pay to Chez Panisse Restaurant. Because Chez Panisse developed several of the Elmwood recipes (about half of the orders served at each meal), Elmwood must pay Chez Panisse $0.20 for each meal they serve using a Chez Panisse recipe. Overhead costs are allocated equally across Breakfast, Lunch and Dinner because Elmwod is open 4 hours a day for each meal.

Should Elmwood stop serving breakfast? If they do, by how much would their profits change? Explain you answer with respect to all the costs Elmwood faces.

MULTI-PART QUESTIONS

Question 4 [35 points]

The Oregon Shakespeare Festival in Ashland has been heavily affected by smoke from the recent wildfires in Oregon and California.[1] Bill Rauch, the festival's artistic director, said in an interview that the financial losses are a result of over 20 performances of the three outdoor main-stage shows having been either canceled or moved to a smaller indoor theater, which has diminished ticket revenue.

A team at the festival meets each morning to assess the air quality and the forecast for the day to determine whether to ***cancel or move a performance indoors***, Mr. Rauch said. If it seems as if the air quality will be poor, that night's outdoor performances are moved indoors. But if there is a chance of good air quality, organizers wait until the evening to decide. Shows are sometimes canceled at the last minute if air quality deteriorates during the day.

If the decision is made early in the day to move the show indoors, then revenues in the smaller venue will be $36,000. If instead the team decides to wait, then moving the show indoors will not be possible, and the show will either be canceled or not. If it is cancelled, the revenues are $0, while if it runs in the large outdoor venue then revenues are $80,000.

a. [5] Imagine a morning when the likelihood of the outdoor show being cancelled is 0.5. If Mr. Rauch wishes to maximize expected revenues, what decision should be made in the morning? What would the expected profit be? Draw a decision tree.

[1] See Peter Libbey's article "Wildfire Smoke Disrupts Oregon Shakespeare Festival" *New York Times,* August 24, 2018.

b. [5] On that morning Mr. Rauch has chosen to move the show indoors. What, if anything, can you conclude about Mr. Rauch's risk preferences? Explain why.

It turns out that festival goers may dislike the smaller indoor venue, depending on their tastes for indoor versus outdoor venues. The tastes of theatre goers for indoor venues are not correlated with the weather. With probability 1/3 the demand for the indoor venue will be high and revenues from moving the show indoor will be $48,000 while with probability 2/3 the demand will be low and revenues will be $30,000.

c. [10] Consider that morning when the likelihood of smoke causing a cancelation is 0.5. If Mr. Rauch is risk neutral and maximizes expected profits, which decision should he make?

d. [5] If Mr. Rauch is extremely risk averse, can you rationalize a different decision than you found in part c. above? Explain why.

e. [10] Assume Mr. Rauch is risk neutral and his assessment of the demand for indoor venues is correct (high/low with probabilities 1/3 and 2/3 respectively). Ms. Jones, a successful market researcher, knows the actual realization of the demand for indoor venues before Mr. Rauch must make his decision in the morning. How much is this information worth to Mr. Rauch? Explain.

Question 5 [30 points]

You own monopoly rights to produce Qubits. The market for these has proved to be surprisingly strong. Given a price p, annual quantity demand in the U.S. is given by $Q_U = 305 - (1/2)P$ while annual quantity in the rest of the world is given by $Q_W = 1{,}220 - 2P$. You can only sell Qubits at a single uniform price-per-unit that applies in any market you are in (i.e., the same price in all countries.) You can build Qubits in plants that you can lease for $30,000 per year. Each plant has a maximum annual capacity of 100 Qubits, but you can lease as many plants as you want. Once leased, the plants have no other uses for a full year, they cannot be sub-leased, and the lease cannot be broken or terminated before the end of the one-year term. Your direct costs total $10 per Qubit.

a. [10] At what price do you sell Qubits? How many do you sell in each market (U.S. and world)? How many plants do you lease?

b. [10] Suppose that after you had leased (and paid for) the number of plants you chose in part (a.) above, but before yearly production begins, the U.S. federal government passes regulation that does not allow the sale of Qubits in the U.S. Now how many Qubits do you sell and at what price?

c. [10] The first year has passed and the U.S. sticks to the regulation that bans the sale of Qubits in the U.S. Now you can decide if you want to keep the same number of plants or if you want to lease a different amount. How many plants do you lease? What are the price and quantity sold and what is your yearly profit?